4.4 Factoring Polynomials

Name _____ Period ____ Date _____

Algebra II 4.4 (30-minute plan) · Student sheet

I can…

- factor polynomials by factoring out a monomial and by grouping.
- factor sums and differences of cubes and polynomials in quadratic form.
- use the Factor Theorem.

Vocabulary · fill in the missing word

Factored completely — written as a product of _____ polynomials with integer coefficients

Sum of two cubes — $a^3 + b^3$ = _____

Difference of two cubes — $a^3 - b^3$ = _____

Quadratic form — $au^2 + bu + c$, where u is an _____ such as x^2

Factor Theorem — $x - k$ is a factor of $f(x)$ _____ when $f(k) = 0$

Part 1 · Notes and examples (fill in each note, then work the example)

Sum of two cubes: $a^3 + b^3$ = _____

Difference of two cubes: $a^3 - b^3$ = _____

Example 1

Factor completely.

a. $x^3 + 27$ b. $8x^3 - 1$

Factor Theorem: Factor Theorem: $x - k$ is a factor exactly when _____.

Example 2

Factor completely: $x^3 - 3x^2 - 4x + 12$

Part 2 · Partner practice

Partner practice

Factor $x^3 - 64$.